Redux

Redux is a predictable state container for JavaScript apps, mainly used with React but also usable elsewhere. It helps manage global state in a structured and predictable way.

1. **Store – a single source of truth.**
   Add all slices as key value pair

2. **Actions – plain objects that describe what happened.**

   ```
   { type: "INCREMENT" }
   ```

   The event triggered

3. **Reducers – pure functions that define how state changes based on actions.**

   ```
   const counterReducer = (state = 0, action) => {
    switch(action.type) {
      case "INCREMENT": return state + 1;
      case "DECREMENT": return state - 1;
      default: return state;
    }
   }
   ```

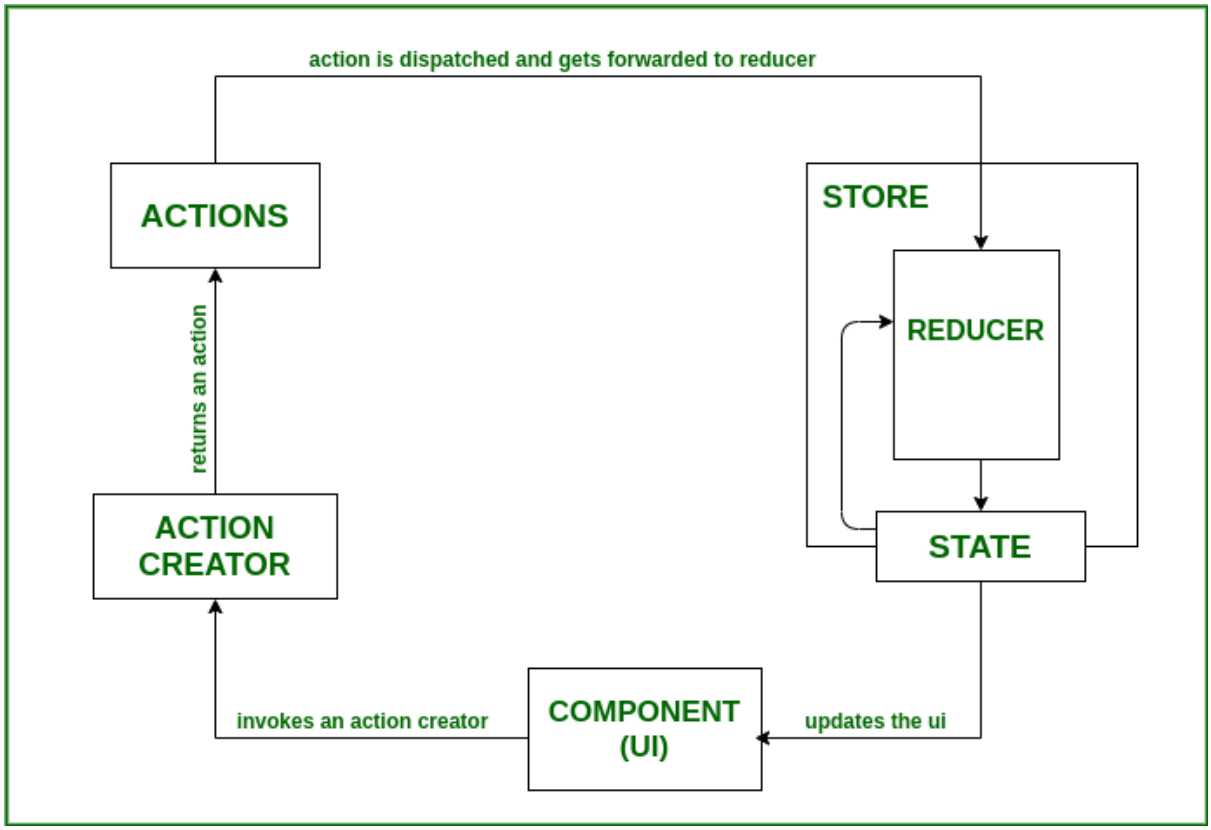
   function called by action

4. **Dispatch – a function to send actions to the store.**

   ```
   store.dispatch({ type: "INCREMENT" });
   ```

   event handling with dispatch triggering action

5. **Selectors – functions to select a slice of state (optional but recommended).**



2. Why Redux?

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How Context API Rendering Works

- **Context API triggers a re-render in all consuming components whenever the value of the provider changes, even if a component only needs part of the value.**
- **Example:**

```
<CounterContext.Provider value={{ count, increment, decrement }}>
  <ComponentA />   // uses count
  <ComponentB />   // only displays a label, doesn't use count
</CounterContext.Provider>
```

- **Problem: When count changes, both ComponentA and ComponentB re-render, even if ComponentB doesn't use count.**
- **This happens because React cannot know which part of the context value a component actually depends on — any change to the object triggers a re-render for all consumers.**

Why This Can Hurt Performance

- **In a small app, it's fine.**
- **In a large app with many components, frequent updates in context cause unnecessary re-renders, which can slow down the UI.**
- **Example: Imagine count updates 60 times per second — every consuming component will re-render.**

2. How Redux Solves This

- **Both Context API and Redux provide global state storage.**
- **Context API: every component that consumes the context re-renders whenever the context value changes, even if the part of the state the component cares about hasn't changed.**
- **Redux: each component subscribes only to the slice of state it actually uses (via useSelector or connect).**

  `const count = useSelector(state => state.counter.count);`

  - **If another slice updates, your component won't re-render.**
  - **Only components using the updated slice re-render.**

✅ **Result: Redux gives fine-grained control over re-renders, which improves performance in larger, frequently-updating apps.**

Problems Redux solves:

1. **Predictability**
    - **State changes are centralized and happen in a predictable manner.**
    - **Every change is traceable via actions, making debugging easier.**
2. **Single Source of Truth**
    - **One store for the whole app → easier to understand and debug.**
3. **Ease of Testing**
    - **Reducers are pure functions → easy to test independently.**
4. **Debugging Tools**
    - **Redux DevTools let you see every action and state change.**
    - **You can time travel: go back and forward between states.**
5. **Middleware Support**
    - **Intercept actions for logging, async calls, error handling, analytics, etc.**
    - **Examples: redux-thunk, redux-saga.**
6. **Decoupling**
    - **Components don't directly manipulate global state; they dispatch actions.**

Redux in React

Step 0: Setup

1. **Make a new React app:**

   npx create-react-app redux-counter
   cd redux-counter

2. **Install Redux and React-Redux:**

   npm install @reduxjs/toolkit react-redux

@reduxjs/toolkit → modern way to write Redux without boilerplate.

react-redux → connects React to Redux store.

Step 1: Create a Redux Slice

What is a slice?
A slice is a single piece of state and all its logic (reducers + actions) combined.
Redux Toolkit makes it super simple with createSlice.

1. **Create folder: src/redux/**
2. **Create file: counterSlice.js**

```
// src/redux/counterSlice.js
import { createSlice } from "@reduxjs/toolkit";

// Step 1: Define initial state
const initialState = {
  count: 0
};

// Step 2: Create slice
export const counterSlice = createSlice({
  name: "counter",       // slice name
  initialState,          // starting state
  reducers: {            // functions to handle actions
    increment: state => {
```

```
    state.count += 1;   // Immer library allows "mutating" state safely
  },
  decrement: state => {
    state.count -= 1;
  },
  reset: state => {
    state.count = 0;
  }
 }
});
```

```
// Step 3: Export actions to use in components
export const { increment, decrement, reset } = counterSlice.actions;

// Step 4: Export reducer to add to store
export default counterSlice.reducer;
```

Explanation of steps:

- **Step 1: initialState → the default state of this slice.**
- **Step 2: createSlice → automatically creates action types and reducer for us.**
- **Step 3: Export actions (increment, decrement, reset) → components will call these to change state.**
- **Step 4: Export reducer → will go into the store.**

Step 2: Configure the Redux Store

What is the store?
The store is the single source of truth for your app's state. All slices are combined here.

Create store.js in src/redux/:

```
// src/redux/store.js
import { configureStore } from "@reduxjs/toolkit";
import counterReducer from "./counterSlice";

// Step 1: Configure store
export const store = configureStore({
  reducer: {
    counter: counterReducer // adding our slice reducer
  }
});

// Redux DevTools enabled by default
```

Explanation:

- **configureStore → modern way to create store, no deprecated createStore.**
- **reducer → combine slices here. Right now we only have counter.**
- **Redux DevTools and middleware are automatically configured. ✅**

Step 3: Connect Redux to React

Wrap the whole app in a <Provider> so all components can access the store.

Edit src/main.js:

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```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import { Provider } from "react-redux"; // step 1
import { store } from "./redux/store";   // step 2

// Step 3: Wrap app with Provider
const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(
  <Provider store={store}>
    <App />
  </Provider>
);
```

Explanation:

- **Provider → makes the Redux store available to every component inside <App>.**
- **Without this, components cannot access the state.**

Step 4: Create Counter Component

We now create the UI and connect it to Redux.

```
// src/Counter.js
import React from "react";
import { useSelector, useDispatch } from "react-redux";
import { increment, decrement, reset } from "./redux/counterSlice";

const Counter = () => {
  // Step 1: Get state from store
  const count = useSelector(state => state.counter.count);

  // Step 2: Get dispatch function
  const dispatch = useDispatch();

  return (
    <div style={{ textAlign: "center", marginTop: "50px" }}>
      <h1>Counter: {count}</h1>
      <button onClick={() => dispatch(increment())}>+</button>
      <button onClick={() => dispatch(decrement())}>-</button>
      <button onClick={() => dispatch(reset())}>Reset</button>
    </div>
  );
};
```

to pass args in dispatch

```
dispatch(increment(3))
```

in slice action

```
increment=(state,action)=>{
        data=action.payload; //3
}
export default Counter;
```

Step-by-step explanation:

1. **useSelector → subscribes to store state and gets the count.**
2. **useDispatch → returns dispatch function to send actions to the store.**
3. **Dispatch actions (increment, decrement, reset) → reducer updates the state → UI automatically re-renders.**